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MILITARY SPECIFICATION

POWER SUPPLY, CATHODIC PROTECTION,

AUTOMATIC CONSTANT CURRENT TYPE, INACTIVE FLEET

1. SCOPE

1.1 Scope. This specification covers the requirements for automatically controlled rectifiers of the constant current type for the cathodic protection of inactive ships. The direct current (d.c.) output of the rectifier is automatically maintained at a set value by saturable reactor control of output voltage to compensate for variations in environmental parameters.

1.2 Classification. Power supplies shall be class R of MIL-P-15736 and of the following types as specified (see 6.1):

Type I - 50 volt - 50 ampere maximum output
Type II - 50 volt - 100 ampere maximum output

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
QQ-A-250 - Aluminum and Aluminum Alloy Plate and Sheet, General Specification for.
QQ-S-775 - Steel Sheets, Carbon, Zinc-Coated.

MILITARY
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Power supplies shall conform to the requirements of class R power supplies of MIL-P-15736 except that qualification does not apply and as specified hereinafter. (Where any conflict exists between MIL-P-15736 and this specification, the latter shall govern.)

3.2 Electrical ratings. Electrical ratings shall be as specified in table I.

Table I - Electrical ratings.

Parameter	Requirements
Input power:	
Alternating current (a.c.)	440 volts
Frequency (input)	60 hertz
Phase (input)	Single phase
D.c. volts (output)	See 1.2 and 3.4.1
D.c. amperes (output)	See 1.2 and 3.4.1
Duty	Continuous
Type of load	Cathodic protection (resistive load)
D.c. output ammeter (minimum range)	0-120 percent of rated d.c. current
D.c. output voltmeter	0-50 volts

FSC 6130

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Reverse polarity and surge voltage protected (output voltage held to ± 1.0 percent AC voltage variation)

3.3 A.c. input.

3.3.1 Rectifiers. Rectification of the output of the power transformer shall be accomplished by means of a combination of one or more rectifier stacks or assemblies in a full wave arrangement. The use of combinations of standard size rectifying devices is encouraged, so that, for a given manufacturer, the number of different kinds of rectifying devices used shall be as small as possible.

3.3.2 Rectifying agent. The rectifying agent shall be silicon diodes.

3.4 D.c. output. The d.c. output voltage of the power supply shall be controlled by a saturable reactor so that the pre-set output current is maintained regardless of change in the load impedance.

3.4.1 Output control. The automatic control shall be provided by a saturable reactor and control assembly which continuously monitors the output current and varies the voltage to the rectifier stack(s) so that output current is maintained at the value set on the control rheostat. The saturable reactor, at any current setting, shall provide continuous control of the output voltage over a range of 5 to 100 percent of the rated output (see 1.2). The sensitivity of the saturable reactor and control assembly shall be such that the current output is controlled to within plus or minus 10 percent of the required current output which shall be set on the control rheostat.

3.5 Electrical connections. The input and output electrical connections shall be brass stud type terminals. The output terminals shall be accessible without opening the enclosure, but shall be protected against weather and inadvertent contact by personnel. Minimum terminal size shall be determined by the maximum current through the connection.

3.5.1 Input. A terminal board shall be provided with all terminals clearly marked showing connection points for 440 volt input leads.

3.5.2 Output. Power supplies shall be provided with terminals for convenient connection of output load leads. At least 6 ship ground (negative) terminals on a common bus and 6 anode terminals shall be provided. To accommodate varying cable lengths, voltage drops and hull bottom conditions for multiple ship arrangements, the anode terminals will be divided into 3 groups of 2 terminals which simultaneously provide output levels at 100, 66, and 33 percent of the output voltage of the saturable reactor. Each level shall be capable of carrying the full current output.

3.6 Overload protection. The saturable reactor control circuit shall be such that overloading by a short circuit across the output will not interrupt the output current, but will limit the current to less than 120 percent of the full rated value. The rectifier shall be capable of operating indefinitely without harm at a short circuit condition. No fuses shall be used. Circuit breakers shall be permitted only at the 440 volt a.c. input.

3.7 Ambient temperature. The operating ambient temperature range shall be minus 20 to plus 50°C.

3.8 Ventilation. Ventilation shall be natural draft.

3.9 Enclosure. The enclosure shall be dripproof protected with a hinged front access door in accordance with MIL-P-15736. The "on-off" switch, control rheostat and meters shall be flush mounted on a rigid front member of the enclosure in an area protected from mechanical damage. All openings in the enclosure, other than the door, shall be screened to prevent the entrance of insects. The enclosure shall be fabricated from steel sheet conforming to QQ-S-775 or aluminum sheet conforming to QQ-A-250.

3.10 Size and weight. The maximum size shall be 24 cubic feet and the maximum weight shall be 650 pounds.

3.11 Components. All electronic components in the control circuits, except for those specified in 3.11.1 and 3.11.2 shall be magnetic amplifier or solid state devices.

3.11.1 Meters. Separate meters shall be provided for indicating the output voltage and load current of the power supply. Meters shall be 3-1/2 inches round panel type in accordance with MIL-M-10304. The meter full scale range shall be as indicated in table I. Meters shall be hermetically sealed, and shall be provided with an external "zero adjust" device. The meters shall be mounted on the power supply enclosure exterior so that they may be read without opening the enclosure door.

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3.11.2 Miscellaneous controls. The following miscellaneous controls shall be provided:

(a) A locking knob type rheostat connected to the control assembly to permit a smooth continuous adjustment of the output current, as required, over the specified range.
(b) A switch shall be provided for on-off operation of the power supply at the a.c. input line. The switch shall disconnect both sides of the a.c. input.
(c) A rotary selector switch to permit the d.c. output voltmeter to measure the voltage on any of the three output voltage levels.
(d) Surge suppressors shall be provided on the a.c. input and the d.c. output to provide protection from lightning.

3.12 Repair parts. The following repair parts shall be supplied for each power supply in accordance with MIL-P-15736.

(a) One silicon diode of each type used for power rectification.
(b) One surge suppressor of each type used.

3.13 Drawings. Drawings shall be in accordance with MIL-P-15736.

3.13.1 A permanent laminated plastic schematic indicating the wiring arrangement of electrical components shall be attached to the inside of the enclosure door.

3.14 Technical Manuals. Technical manuals shall be in accordance with MIL-P-15736.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance provisions shall be in accordance with MIL-P-15736, except for qualification which is not applicable to this equipment and the test of 4.3 herein. Prior to government acceptance of the product, the manufacturer shall submit to the procuring activity the master drawings for the product, other specific information requested by the procuring activity, and test evidence that the product conforms to all requirements of MIL-P-15736 including environmental requirements, except for shock and vibration. Environmental tests are required only once on any specific design of equipment. Any change in the product will require complete testing against all requirements.

4.3 Tests.

4.3.1 Voltage regulation. The power supply shall be tested for steady state voltage regulation performance in accordance with the procedures of MIL-P-15736, except as modified in 4.3.1.1 herein.

4.3.1.1 Steady state voltage regulation shall be measured at 4 equally spaced steps of output current and at resistive loads requiring from 5 percent of rated voltage to full rated load in approximately 5 equally spaced steps and overloaded to a short circuit condition.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-P-15736.

6. NOTES

6.1 The notes specified in MIL-P-15736 and the following are applicable to this specification.

(a) Type of rectifier required (see 1.2).

Preparing activity:
Navy - SH
(Project 6130-N086)

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